

Pulkit Sharma

Third Year Undergraduate
Department of Chemical Engineering

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech	Indian Institute of Technology Kanpur	7.1/10
2020	Class XII (CBSE)	Father Agnel School, Delhi	94.2%
2018	Class X (CBSE)	Father Agnel School, Delhi	94.2%

Scholastic Achievements

- Secured All India Rank **3850** in **Joint Entrance Examination Advanced 2021** among shortlisted **1.5 Lakh** candidates
- Software Development Engineer Intern at **Amazon**

Key Projects

- Introduction to Data Science in Chemical Engineering** | *Chemineers Society, IIT Kanpur* (Dec'22-Jan'23)
 - Implemented a **linear regression model** using the **Sklearn** library and compared the results with the help of **scratch**
 - Trained a **polynomial Regression model** with different **Loss functions** using libraries Numpy, pandas and matplotlib
 - Created an **Artificial Neural Network** from scratch using a data-set of **9568 data points** and obtained **88%** accuracy
 - Divided the data-set** into training, validation, and testing, then implemented a **Neural Network** with hidden layers
- Computational Project** | *Course Project - ESO208A* (Aug'22-Nov'22)
 - Implemented **Romberg Integration** using Gauss-Legendre Quadrature and generated graphs to visualize the data set
 - Solved Initial Value Problems using **EULER Method** and second and fourth order **RUNGE KUTTA Methods**
 - Computed **linear** and **non linear** equations using **CROUT, DOLITTE, CHOLSKY** methods on **MATLAB**
- Introduction to 3D** | *Summer Project, SnT Council, IIT Kanpur* (May'22-Jul'22)
 - Proficiently used **Blender**, a leading **3D Graphic Modeling** software, to create visually appealing 3D models
 - Rendered numerous 3D models and utilized **Node Creation** in Blender to enhance their visual quality and realism
 - Implemented concepts like modelling, texturing, shading and rigging characters to understand 3D digital art
- Introduction to Multiphase Flow** | *Chemineers Society, IIT Kanpur* (Feb'23-Jun'23)
 - Studied the basic concepts of Heat Transmission, Heat Exchangers, Fouling, Surface Tension and Multiphase Flow
 - Performed a comparative analysis between **Analytical and Numerical solutions** of the **Hagen Poiseuille Equation**
 - Derived the **Critical Insulation Thickness** for a sphere using the **Conduction Equation** for spherical coordinates
 - Understood the significance of using Logarithmic Mean Temperature Difference (**LMTD**) in optimizing thermal processes
- Manufacturing Working Model of Sugarcane Juice Extractor** | *Course Project - TA202A* (Aug'22-Nov'22)
 - Collaborated** as a team of **7 students** to develop an **efficient** as well as **user-friendly** sugarcane juice extractor
 - Fabricated parts using **lathe machining, turning, milling, welding** operations and other manufacturing techniques
 - Prepared comprehensive documentation including technical drawings using **AutoCAD**, and assembly instructions
 - Estimated the total cost of the working model around **Rs.12,500** by allocating value to each material used in the project

Positions of Responsibility

- COMPANY COORDINATOR** | *Students' Placement Office, IITK* (Jul'22-Jul'23)
 - Acted as a **Main Point of Contact** for various companies during internship and placement season 2022-23
 - Coordinated** in a structured team of **150+ members** for handling the recruitment process of **1200+ Candidates**
 - Recorded one of the highest number of offers rolled out in placements as well as internships among other IITs
- SECRETARY** | *Game Development Club, SnT Council, IIT Kanpur* (Sep'22-Jul'23)
 - Helped in an organized conduction of **SnT Pavilion** where we gave the freshers a fair idea of what the club does
 - Started '**Just Insights**' social media campaign with an audience of **2500+** to promote GameDev within the institution
 - Actively contributed towards the club's ongoing **Render Monday** series spread across multiple social media platforms
 - Mentored a project **Game Art and FX** under **SnT Council** to encourage freshers to participate in the club's activities
- STUDENT GUIDE** | *Counselling Service, IIT Kanpur* (Oct'22-Present)
 - Responsible for ensuring a smooth transition of **5 freshmen** into college, helping them acclimatizing to IIT Kanpur
 - Assisted the members of the Core team while supervising the registration and orientation of **1000+** students
 - Conducted weekly interactive sessions with the mentees to help them with their problems and guide them academically

Technical Skills

- Programming Languages and Libraries:** C, C++, MATLAB, Python, Numpy, Pandas, Matplotlib, LaTeX
- Software:** HTML, Arduino, AutoCAD, Canva, Adobe Photoshop, Adobe Illustrator, Blender, Unity, Aseprite, Dragonbones

RELEVANT COURSES

Fundamental of Computing Linear algebra and Differential Equation Introduction to Thermodynamics Chemical Engineering Thermodynamics	Computational Methods in Engineering Introduction to Electronics Introduction to Chemical Engineering Fluid Mechanics and its Application
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Extra Curricular

- Secured **2nd position** in a task by Game Development Club in **Takneek'22** (Inter-Hall Science and Technology competition)
- Performed in a group dance on Freshers' Night'21 hosted by the **Media and Culture Council**, Students Gymkhana
- Performed **Nukkad** (Street play) and represented pool Kshatriyas in **Galaxy'22** (Inter-Hall Cultural competition)
- Volunteered for donating blood in a blood donation camp organized by **Raktarpan, IITK** and **IMA, Kanpur**